

UNDERSTANDING ON REAL - WORLD SCENARIOS

Supervised: Regression (Predict Continuous Numbers)

Algorithm	When to Use	Example Problem	Example Input	Example Output
Linear Regression	Baseline, linear relationship, interpretability needed	Predict house price based on square footage	1500 sq ft	\$300,000
Polynomial Regression	Curved relationship (one feature, non-linear)	Predict acceleration based on car speed	50 mph	3.2 m/s ²
Logistic Regression	Binary classification (despite name)	Will customer click ad?	Age=25, Time=10s	Yes (1) or No (0)
Ridge Regression	Linear regression with overfitting, many small coefficients	Predict stock price from 200 economic indicators	200 features	\$152.30
Lasso Regression	Need feature selection (many irrelevant features)	Predict diabetes progression from 1000 genetic markers	1000 SNPs	85.2 (progression score)
Elastic Net	Correlated features + feature selection needed	Predict housing price from size, bedrooms, location, age	50 features (correlated)	\$450,000
Stepwise Regression	⚠️ Rarely use (outdated)	Automated model selection (historical)	20 candidate variables	Best subset of 5 variables

Supervised: Classification (Predict Categories)

Algorithm	When to Use	Example Problem	Example Input	Example Output
Naive Bayes	Text classification, small data, independent features	Classify email as Work, Personal, or Spam	"Meeting at 2pm"	Work
K-Nearest Neighbors (KNN)	Small dataset, no training time, intuitive boundaries	Recommend movie based on similar users	User age=25, rating=4.5	Sci-Fi
SVM (Support Vector Machines)	Small/medium data, clear margin, high dimensions	Classify handwritten digits (0-9)	8x8 pixel image (64 features)	7
Decision Tree	Need interpretability, mixed data types	Approve or deny loan application	Income=\$50k, Credit=700	Approved
Random Forest	Tabular data, robust without tuning, feature importance needed	Predict customer churn	Usage=20hrs, SupportCalls=3	Churn
XGBoost	Highest accuracy needed, tabular data, competitions	Predict fraud transaction	Amount=\$1000, Location=NYC	Fraud

Unsupervised Learning (No Labels)

Algorithm	When to Use	Example Problem	Example Input	Example Output
K-Means Clustering	Spherical clusters, equal size, known K	Segment supermarket customers	Spending=\$500, Frequency=10x/month	Cluster 1 (High value)
DBSCAN	Irregular shapes, noise, unknown number of clusters	Find star clusters in astronomy	X,Y coordinates of 10,000 stars	3 clusters + noise
PCA (Principal Component Analysis)	Reduce features, remove correlation, visualization	Compress 1000 gene expressions to 2D for plotting	1000 gene values	PC1=2.5, PC2=-1.3

Evaluation & Validation			
Technique	When to Use	Example Scenario	What It Reveals
Precision	When false positives are costly	Spam detection (don't mark good email as spam)	Of emails marked spam, % actually spam
Recall	When false negatives are costly	Cancer detection (don't miss a tumor)	Of actual cancers, % detected
F1 Score	Need balance between precision & recall	General classification (no strong preference)	Harmonic mean of precision & recall
Residual Plots	After regression to check assumptions	Linear regression on house prices	Non-random pattern = missing non-linearity
Grid Search	When you have < 10 hyperparameter combinations	Tuning Random Forest (n_estimators, max_depth)	Best parameter combo
K-Fold CV	Limited data (can't afford single validation set)	1000 samples, need reliable accuracy estimate	Average accuracy across 5 folds
Learning Curve	Diagnose underfitting vs overfitting	Model performing poorly on validation	High train error = underfitting
ROC Curve	Binary classification with threshold tuning	Fraud detection (balance sensitivity/specificity)	AUC score (0.5=random, 1.0=perfect)
PR Curve	Imbalanced classification (rare class important)	Rare disease detection (1% prevalence)	Better than ROC for imbalanced data

Deep Learning: Neural Networks				
Algorithm	When to Use	Example Problem	Example Input	Example Output
Perceptron	Learning foundation only (not practical)	Linearly separable classification	AND gate inputs (0,0)	0
MLP + Backpropagation	Small/medium non-tabular, <100k samples	Predict customer lifetime value	50 customer features	\$1,200
CNN + Kernels/Pooling	Image data, spatial patterns	Detect pneumonia from chest X-ray	512x512 grayscale image	Pneumonia (1) or Normal (0)
RNN / LSTM	Sequential data, time dependencies	Predict next word in sentence	"The cat sat on the"	"mat"
Dropout	Prevent overfitting in large neural nets	Training CNN on 1M images	Randomly drop 50% of neurons during training	Better generalization
Batch Normalization	Speed up training of deep networks	50-layer ResNet	Normalize each layer's outputs	Faster convergence
Adam Optimizer	Default optimizer for most deep learning	Any neural network training	Combines momentum + RMSprop	Adaptive learning rates

NLP (Natural Language Processing)				
Algorithm/Tech	When to Use	Example Problem	Example Input	Example Output
Bag of Words	Baseline text classification	Spam detection	"Win free money now"	[1,1,1,1] (word counts)
TF-IDF	Important words matter, common words less	Search engine ranking	"How to bake bread"	High weight to "bake","bread"
Word2Vec	Need word meanings as vectors	Find similar words	"king" - "man" + "woman"	"queen"
RNN/LSTM	Sequence modeling for text	Sentiment analysis	"This movie was amazing!"	Positive
Transformer (BERT)	State-of-the-art for most NLP	Question answering	"What is ML?" + context	"Machine learning is..."

Quick Decision Matrix (Most Common Scenarios)

If you have...	Don't use...	Use...	Example
Small tabular data (n<1000)	Deep Learning	Logistic Regression, SVM	500 medical patients
Large tabular data (n>10k)	SVM (too slow)	XGBoost, Random Forest	100k bank transactions
Image data	XGBoost, Random Forest	CNN	50k labeled photos
Text data	SVM (without TF-IDF)	Naïve Bayes → BERT	10k movie reviews
Time series (short)	RNN (overkill)	Linear Regression with lags	500 days of sales
Time series (long, complex)	Simple regression	LSTM	50k hours of sensor data
Need interpretability	XGBoost, Neural Nets	Linear Regression, Decision Tree	Loan approval (legal requirement)
Imbalanced classes (1:100)	Accuracy metric	PR curve + Recall/F1	Fraud detection (1% fraud)

Real-World Example: Complete Decision Path

Problem: Predict if a bank customer will default on a loan.

Step	Question	Answer	Algorithm Choice
1	Labeled?	Yes (default history)	Supervised
2	Output type?	Default (Yes/No)	Classification
3	Data type?	Tabular (income, credit score, debt)	Not images/text
4	Data size?	50,000 rows	Medium-large
5	Need interpretability?	Yes (bank regulations)	Decision Tree or Logistic Regression
6	Imbalanced?	Yes (5% default rate)	Use PR curve, not ROC
Final choice:	Logistic Regression (baseline) + XGBoost (performance) + Decision Tree (interpretability)		

One-Sentence Summary for Each Algorithm

Algorithm	One-Sentence Rule
Linear Regression	"If the relationship looks like a straight line"
Logistic Regression	"If you need probability of yes/no"
Ridge/Lasso	"If linear regression is overfitting"
KNN	"If your data is small and similar points have similar outputs"
SVM	"If you have a clear boundary between classes"
Decision Tree	"If your boss asks 'why did the model decide that?'"
Random Forest	"If you want Decision Tree but better"
XGBoost	"If you want to win a Kaggle competition"
K-Means	"If you know how many groups you want"
DBSCAN	"If you don't know how many groups and there's noise"
PCA	"If you have too many features to see or compute"
CNN	"If your data looks like a picture"
RNN/LSTM	"If order matters (time, sentence, sequence)"
BERT	"If you have lots of text and a GPU"